# Automated Black-Box Detection of Side-Channel Vulnerabilities in Web Applications



## Content

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Peter Chapman and David Evans *18th ACM Conference on Computer and Communications Security* (CCS 2011), Chicago, Illinois 17-21 October 2011

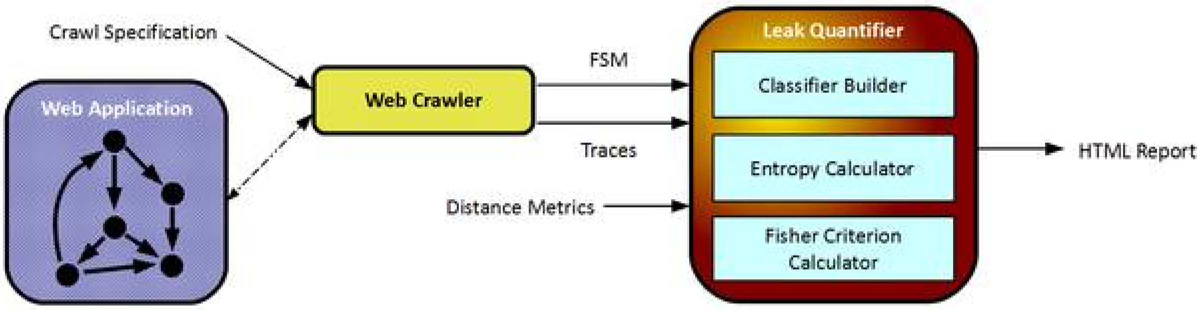
#### Abstract

Web applications divide their state between the client and the server. The frequent and highly dynamic client-server communication that is characteristic of modern web applications leaves them vulnerable to side-channel leaks, even over encrypted connections. We describe a black-box tool for detecting and quantifying the severity of side-channel vulnerabilities by analyzing network traffic over repeated crawls of a web application. By viewing the adversary as a multi-dimensional classifier, we develop a methodology to more thoroughly measure the distinguishably of network traffic for a variety of classification metrics. We evaluate our detection system on several deployed web applications, accounting for proposed client and server-side defenses. Our results illustrate the limitations of entropy measurements used in previous work and show how our new metric based on the Fisher criterion can be used to more robustly reveal side-channels in web applications.

### Paper

Full paper: PDF (12 pages)

## Project Website

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